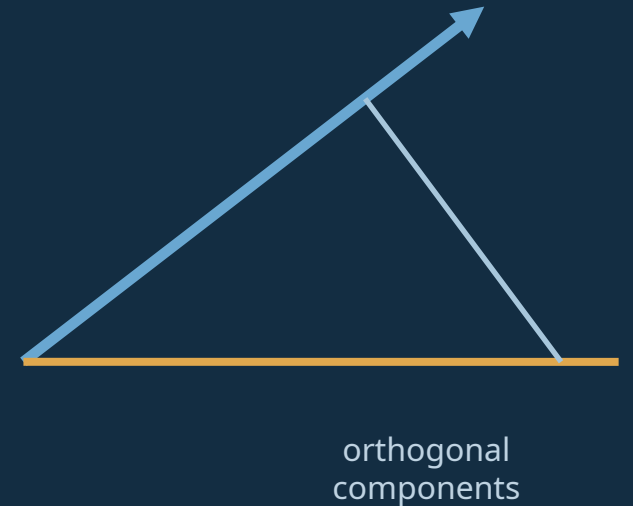


Quadratic Forms and Cochran's Theorem

From squared projections to chi-squared distributions and classical linear-model inference

$$\frac{Y^T (I - H) Y}{\sigma^2} \sim \chi_{n-p}^2$$



Learning objectives

The goal is to make the distributional machinery behind sums of squares feel geometric rather than mysterious.

ROADMAP

- Express a quadratic form as a weighted sum of squared coordinates using the spectral decomposition.
- Recognize when a normal quadratic form has a central or noncentral chi-squared distribution.
- State and interpret Cochran's theorem in both its classical and projection-matrix forms.
- Use orthogonal projection matrices to derive the distributions and independence of regression sums of squares.
- Derive the usual chi-squared and F results for residual sums of squares, nested-model tests, and ANOVA.

The distributional chain

BIG PICTURE

1 Normal vector

Start with $Z \sim N(0, I)$.

2 Projection

A symmetric idempotent matrix keeps some orthogonal directions and removes others.

3 Quadratic form

$Z^T P Z$ is the squared length of the projected normal vector.

4 Chi-squared

The number of retained directions - $\text{rank}(P)$ - becomes the degrees of freedom.

5 Cochran

Orthogonal projections produce independent chi-squared pieces.

01

Quadratic forms

How matrices turn normal vectors into sums of squares.

What is a quadratic form?

QUADRATIC FORMS

For a vector $y \in \mathbb{R}^n$ and an $n \times n$ matrix A ,

$$Q(y) = y^T A y$$

Examples from linear models

Residual sum of squares: $Y^T(I-H)Y$.

Regression sum of squares: $Y^T(H-J)Y$.

General extra sum of squares: $Y^T(H_1-H_0)Y$.

Interpretation

A quadratic form is a scalar built from the vector twice. When A is a projection, it becomes a squared Euclidean length.

Only the symmetric part matters

QUADRATIC FORMS

$$A_s = \frac{A + A^T}{2}, \quad A_a = \frac{A - A^T}{2}$$

Every matrix decomposes into a symmetric and a skew-symmetric part.

$$y^T A y = y^T A_s y + y^T A_a y \quad y^T A_a y = 0$$

Conclusion

$$y^T A y = y^T \left(\frac{A + A^T}{2} \right) y$$

Spectral decomposition reveals the structure

QUADRATIC FORMS

Let A be symmetric. Then $A = Q\Lambda Q^T$

with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Define $u = Q^T y$.

$$y^T A y = u^T \Lambda u = \sum_{i=1}^n \lambda_i u_i^2$$

Key idea

A symmetric quadratic form is a weighted sum of squared coordinates in the eigenvector basis.

Distributional consequence

For a standard normal vector, those rotated coordinates remain independent $N(0, 1)$.

A normal quadratic form is usually a weighted chi-squared sum

DISTRIBUTION

If $Z \sim N_n(0, I)$ and A is symmetric, then $Q^T Z \sim N_n(0, I)$. Therefore

$$Z^T A Z \stackrel{d}{=} \sum_{i=1}^n \lambda_i \chi_{1,i}^2$$

Not generally chi-squared

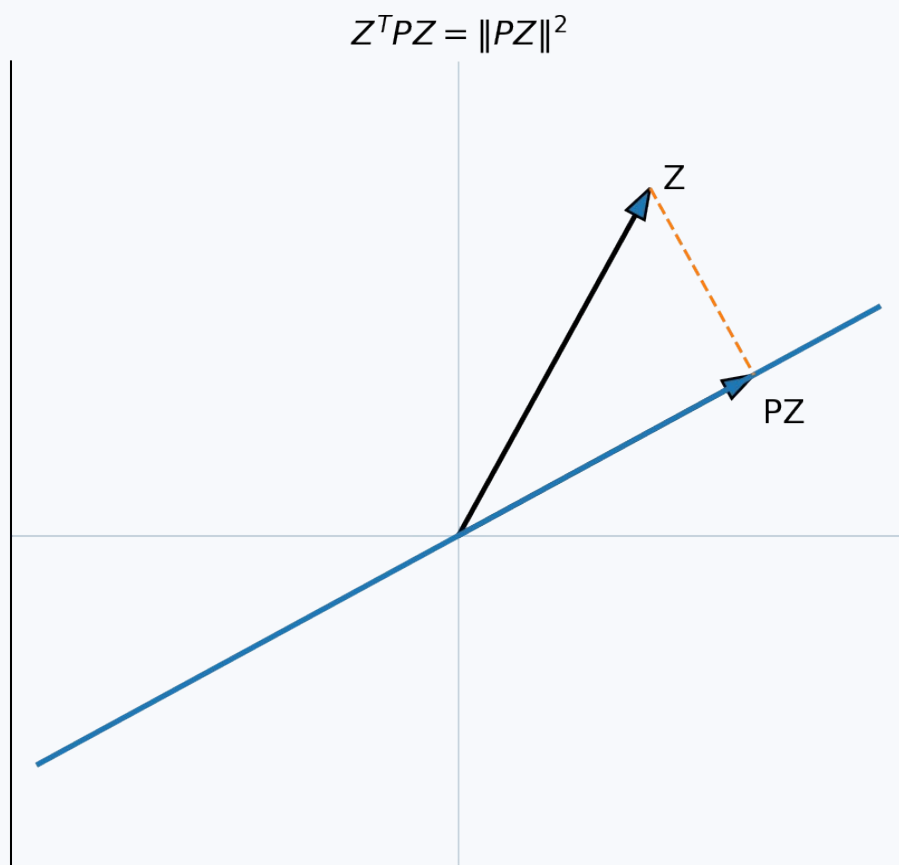
If the nonzero λ_i are arbitrary, the result is a weighted sum of independent χ^2_1 variables.

Exactly chi-squared when...

All nonzero eigenvalues are 1. That is precisely what happens for a symmetric idempotent matrix.

Projection matrices turn quadratic forms into squared lengths

GEOMETRY



$$P = P^T = P^2$$

$$Z^T P Z = (PZ)^T (PZ) = \|PZ\|^2$$

Meaning

P keeps only the component of Z in $C(P)$. The quadratic form measures the squared length in that retained subspace.

Degrees of freedom

The dimension of $C(P)$ is $\text{rank}(P)$, so rank counts the number of independent squared normal coordinates.

Key theorem: idempotent quadratic forms

THEOREM

Central chi-squared result

$$Z \sim N_n(0, I), \quad P = P^T = P^2, \quad r = \text{rank}(P) \Rightarrow \boxed{Z^T P Z \sim \chi_r^2}$$

Condition 1

Normality provides independent standard normal coordinates after an orthogonal rotation.

Condition 2

Symmetric idempotence forces the eigenvalues to be only 0 or 1.

Condition 3

Rank counts the number of 1 eigenvalues and therefore the number of squared normals that remain.

Why the theorem is true

PROOF SKETCH

1

Diagonalize P

Because P is symmetric, $P = Q\Lambda Q^T$ for an orthogonal Q.

2

Use idempotence

$P^2 = P$ implies $\lambda_i^2 = \lambda_i$, so each λ_i is 0 or 1.

3

Rotate the normal vector

$U = Q^T Z$ still has distribution $N(0, I)$.

4

Count the retained coordinates

If $\text{rank}(P)=r$, exactly r eigenvalues equal 1.

$$Z^T P Z = U^T \Lambda U = U_1^2 + \cdots + U_r^2 \sim \chi_r^2$$

Why rank becomes degrees of freedom

INTERPRETATION

Suppose P projects $\mathbb{R}^n$ onto an r -dimensional subspace.

U_1^2

U_2^2

U_3^2

U_4^2

0

0

0

Each retained orthogonal direction contributes one independent squared $N(0,1)$.

So...

$$\text{degrees of freedom} = \#\{\lambda_i = 1\} = \text{rank}(P) = r$$

Noncentral chi-squared forms

DISTRIBUTION

Shifted normal vector

$$Y \sim N_n(\mu, \sigma^2 I), \quad P = P^T = P^2, \quad r = \text{rank}(P) \Rightarrow \boxed{\frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda)}, \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}$$

Central case

If $P\mu = 0$, then $\lambda = 0$ and the distribution is central χ_r^2 .

Alternative hypothesis

If $P\mu \neq 0$, the mean has a component in the tested subspace and the sum of squares becomes noncentral.

Moments of a normal quadratic form

USEFUL FORMULAS

For $Y \sim N_n(\mu, \Sigma)$ and symmetric A ,

Expected value

$$E(Y^T A Y) = \text{tr}(A \Sigma) + \mu^T A \mu$$

Variance

$$\text{Var}(Y^T A Y) = 2 \text{tr}(A \Sigma A \Sigma) + 4 \mu^T A \Sigma A \mu$$

Projection special case

If $Y \sim N(0, \sigma^2 I)$ and P is a rank- r projection, $E(Y^T P Y) = r \sigma^2$ and $\text{Var}(Y^T P Y) = 2r \sigma^4$.

Worked example: a rank-2 projection

EXAMPLE

$$Z^T P Z = Z_1^2 + Z_2^2$$

Matrix properties

P is symmetric and idempotent.
 $\text{rank}(P)=2$.

Distribution

Therefore $Z^T P Z \sim \chi^2_2$.

The third coordinate contributes nothing because P annihilates that direction.

Concept check: when is the form chi-squared?

CHECK

Let $Z \sim N_n(0, I)$. Decide what you can say about $Z^T A Z$.

A symmetric, eigenvalues 2 and 3

weighted sum: $2\chi^2_1 + 3\chi^2_1$

A symmetric idempotent, rank 5

χ^2_5

$A = I$

χ^2_n

A projection, but applied to $\mu \neq 0$

noncentral χ^2 unless $A\mu = 0$

02

Cochran's theorem

A decomposition of total squared normal variation into independent chi-squared pieces.

Why another theorem?

MOTIVATION

The quadratic-form theorem tells us the distribution of one projected sum of squares.

Cochran's theorem adds the crucial second ingredient:

Distribution

Each orthogonal sum-of-squares piece has a chi-squared distribution.

Independence

Different orthogonal pieces are mutually independent.

That independence is what lets us form exact F statistics.

Cochran's theorem: classical quadratic-form statement

THEOREM

Let $Z \sim N_n(0, I)$, and suppose

$$Z^T Z = Q_1 + \cdots + Q_k$$

where each $Q_j = Z^T A_j Z$ is a nonnegative quadratic form of rank r_j .

Rank condition

$$r_1 + \cdots + r_k = n$$

Then

$Q_1, \dots, Q_k$ are mutually independent, and $Q_j \sim \chi^2_{\{r_j\}}$.

The rank condition says that the pieces account for all n orthogonal directions - with no overlap and no missing direction.

Cochran's theorem: the projection-matrix form

LINEAR-MODEL FORM

Suppose

$$P_j = P_j^T = P_j^2, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_{j=1}^k P_j = I$$

For $Z \sim N_n(0, I)$, define $Q_j = Z^T P_j Z$.

$$Q_j \sim \chi_{r_j}^2 \text{ independently,} \quad r_j = \text{rank}(P_j)$$

This is the version to remember

Linear model sums of squares are almost always quadratic forms in mutually orthogonal projection matrices.

Geometric interpretation

GEOMETRY

An orthogonal decomposition of $\mathbb{R}^n$:

S_1

dim = r_1

S_2

dim = r_2

S_3

dim = r_3

S_k

dim = r_k

$$\mathbb{R}^n = S_1 \oplus S_2 \oplus \cdots \oplus S_k$$

P_j projects Z into S_j , and $Q_j = \|P_j Z\|^2$. Orthogonal Gaussian components are independent.

Why Cochran's theorem works

PROOF SKETCH

1

Choose an orthonormal basis

Build a basis adapted to the mutually orthogonal subspaces $C(P_1), \dots, C(P_k)$.

2

Rotate Z

The coordinates $U = Q^T Z$ remain independent $N(0,1)$.

3

Group coordinates by subspace

Each Q_j becomes a sum of squares of a disjoint set of U coordinates.

4

Use disjointness

Disjoint sets of independent normal coordinates produce independent chi-squared sums.

Rank accounting

$$r_1 + \dots + r_k = n$$

Classical example: the sample variance

EXAMPLE

Let $Y_1, \dots, Y_n$ be iid $N(\mu, \sigma^2)$, and define

$$J = \frac{1}{n} \mathbf{1}\mathbf{1}^T, \quad M = I - J$$

Projection onto the mean

J projects onto $\text{span}(\mathbf{1})$, so $\text{rank}(J)=1$.

Projection onto deviations

M projects onto $\text{span}(\mathbf{1})^\perp$, so $\text{rank}(M)=n-1$.

$$Y^T M Y = \sum_{i=1}^n (Y_i - \bar{Y})^2 = (n-1)S^2$$

Distribution

$$(n-1)S^2 / \sigma^2 \sim \chi_{n-1}^2$$

Why the sample mean and sample variance are independent

COCHRAN

$$I = J + (I - J), \quad J(I - J) = 0$$

The two matrices are orthogonal projections of ranks 1 and $n-1$.

Mean component

$JY = \bar{Y}1$ contains the sample mean.

Residual component

$(I-J)Y$ contains the centered deviations and determines S^2 .

Cochran's theorem

$$JY \perp (I - J)Y \implies \bar{Y} \perp S^2$$

The rank condition is not bookkeeping trivia

INSIGHT

In Cochran's theorem, the ranks must add to the dimension of the original normal vector.

Piece 1
rank = r_1

Piece 2
rank = r_2

Piece 3
rank = r_3

Piece k
rank = r_k

$$r_1 + r_2 + \cdots + r_k = n$$

Interpretation

No direction may be counted twice, and no direction may be omitted. In regression, this becomes the familiar addition of degrees of freedom.

03

Application to the normal linear model

Projection matrices convert the geometry of least squares into exact sampling distributions.

Normal linear model setup

LINEAR MODEL

Model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I), \quad \text{rank}(X) = p$$

$$H = X(X^T X)^{-1} X^T, \quad M = I - H$$

H

Orthogonal projection onto $C(X)$.
 $\text{rank}(H)=p$.

M = I - H

Orthogonal projection onto $C(X)^\perp$.
 $\text{rank}(M)=n-p$.

Residuals are a projected normal vector

RESIDUALS

$$e = Y - \hat{Y} = (I - H)Y = MY$$

Because $MX = 0$,

$$e = M(X\beta + \varepsilon) = M\varepsilon$$

Residual distribution

$$e \sim N_n(0, \sigma^2 M)$$

Important subtlety

M is singular: the n residual coordinates are not independent. They live in an $(n-p)$ -dimensional subspace.

Distribution of the residual sum of squares

SSE

$$SSE = e^T e = Y^T M Y$$

Since $MY = M\varepsilon$ and $Z = \varepsilon/\sigma \sim N_n(0, I)$,

$$\frac{SSE}{\sigma^2} = Z^T M Z$$

M is symmetric, idempotent, and $\text{rank}(M) = n - p$.

Therefore

$$\frac{SSE}{\sigma^2} \sim \chi_{n-p}^2$$

Why MSE is unbiased for σ^2

CONSEQUENCE

From the chi-squared law

$$E\left(\frac{SSE}{\sigma^2}\right) = n - p$$

$$E(SSE) = (n - p)\sigma^2$$

Hence

$$MSE = \frac{SSE}{n - p} \quad \text{satisfies} \quad E(MSE) = \sigma^2$$

The residual degrees of freedom are not arbitrary: they are exactly $\text{rank}(I - H)$.

Why $\hat{\beta}$ and the residuals are independent

INDEPENDENCE

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad e = MY$$

Both are linear transformations of the same multivariate normal vector Y .

$$\text{Cov}(\hat{\beta}, e) = (X^T X)^{-1} X^T (\sigma^2 I) M = \sigma^2 (X^T X)^{-1} X^T M = 0$$

Joint normal + zero covariance

$$\hat{\beta} \perp e \implies \hat{\beta} \perp SSE$$

Regression as an orthogonal sum-of-squares decomposition

REGRESSION

Assume the model contains an intercept and define $J = \mathbf{1}\mathbf{1}^T/n$.

$$I - J = (H - J) + (I - H)$$

Corrected total

$$\begin{aligned} \text{SST} &= Y^T(I - J)Y \\ \text{rank} &= n - 1 \end{aligned}$$

Regression

$$\begin{aligned} \text{SSR} &= Y^T(H - J)Y \\ \text{rank} &= p - 1 \end{aligned}$$

Error

$$\begin{aligned} \text{SSE} &= Y^T(I - H)Y \\ \text{rank} &= n - p \end{aligned}$$

$$n - 1 = (p - 1) + (n - p)$$

The regression and error projections are orthogonal

COCHRAN

$$P_R = H - J, \quad P_E = I - H$$

When the intercept is in $C(X)$:

$$P_R^2 = P_R, \quad P_E^2 = P_E, \quad P_R P_E = 0$$

Geometric meaning

The regression component beyond the intercept and the residual component occupy perpendicular subspaces.

Statistical meaning

Under normality, the corresponding sums of squares are independent.

Under the global null, SSR is central chi-squared

GLOBAL F TEST

Suppose the null hypothesis is that all slopes are zero, so $\mu = X\beta$ lies in $\mathcal{C}(J)$.

$$H_0 : \mu \in \mathcal{C}(J)$$

Then $(H-J)\mu = 0$, so the regression quadratic form is central.

By Cochran

$$\frac{SSR}{\sigma^2} \sim \chi_{p-1}^2, \quad \frac{SSE}{\sigma^2} \sim \chi_{n-p}^2, \quad SSR \perp SSE$$

This is the exact distributional foundation of the usual regression F test.

From independent chi-squared variables to the F distribution

F TEST

If $U \sim \chi^2_r$ and $V \sim \chi^2_s$ independently, then

$$\frac{U/r}{V/s} \sim F_{r,s}$$

Apply this to regression:

Global regression F statistic

$$F = \frac{SSR/(p-1)}{SSE/(n-p)} \sim F_{p-1, n-p} \quad \text{under } H_0$$

Nested models: the extra sum of squares

NESTED MODELS

Suppose $C(X_0) \subset C(X_1)$, with ranks $p_0 < p_1$.

$$H_0 = X_0(X_0^T X_0)^{-1} X_0^T, \quad H_1 = X_1(X_1^T X_1)^{-1} X_1^T$$

Extra projection

$$P_{\text{extra}} = H_1 - H_0$$

Key properties

$H_1 - H_0$ is symmetric and idempotent.
 $\text{rank}(H_1 - H_0) = p_1 - p_0$.

$$SSE_0 - SSE_1 = Y^T (H_1 - H_0) Y$$

The reduction in SSE is itself a quadratic form.

Distribution of the extra sum of squares

NESTED MODELS

Under H_0 , the true mean lies in $C(X_0)$, so $(H_1 - H_0)\mu = 0$.

Numerator

$$\frac{SSE_0 - SSE_1}{\sigma^2} \sim \chi_q^2, \quad q = p_1 - p_0$$

Denominator

$$\frac{SSE_1}{\sigma^2} \sim \chi_{n-p_1}^2$$

Cochran supplies independence

The extra-model projection $H_1 - H_0$ is orthogonal to $I - H_1$, so the two chi-squared quantities are independent.

The partial F test falls out immediately

NESTED MODELS

Partial F statistic

$$F = \frac{(SSE_0 - SSE_1)/q}{SSE_1/(n - p_1)} \sim F_{q, n-p_1} \quad \text{under } H_0$$

Numerator

Variation explained by the q additional directions in $C(X_1)$ that are not already in $C(X_0)$.

Denominator

Residual variation left in the orthogonal complement of $C(X_1)$.

The F test is a ratio of mean squared lengths in two independent orthogonal subspaces.

One-way ANOVA is the same geometry

ANOVA

For g groups and n total observations, the usual decomposition is

$$SS_T = SS_{\text{Between}} + SS_{\text{Within}}$$

Between groups

Projection onto group-mean contrasts.
Degrees of freedom: $g-1$.

Within groups

Projection onto residual directions inside groups.
Degrees of freedom: $n-g$.

Under equal means

$$\frac{SS_{\text{Between}} / \sigma^2}{g - 1} \bigg/ \frac{SS_{\text{Within}} / \sigma^2}{n - g} \sim F_{g-1, n-g}$$

Under alternatives: noncentral chi-squared and noncentral F

POWER

If the mean has a nonzero component in the tested subspace, the numerator is no longer central.

For a projection P

$$\frac{Y^T P Y}{\sigma^2} \sim \chi_r^2(\lambda), \quad \lambda = \frac{\mu^T P \mu}{\sigma^2}$$

Interpretation of λ

The noncentrality parameter measures signal strength in the tested subspace relative to noise variance.

This is the bridge from exact null distributions to power calculations.

$$H_0 : \lambda = 0 \quad \longrightarrow \quad H_1 : \lambda > 0$$

A compact distribution table

SUMMARY			
Quantity	Projection	Rank / df	Distribution under null
SSE	$I - H$	$n - p$	$SSE/\sigma^2 \sim \chi^2_{n-p}$
SSR (with intercept)	$H - J$	$p - 1$	$SSR/\sigma^2 \sim \chi^2_{p-1}$
Extra SS	$H_1 - H_0$	$p_1 - p_0$	$\text{Extra SS}/\sigma^2 \sim \chi^2_{q}$
Corrected SST	$I - J$	$n - 1$	$SST/\sigma^2 \sim \chi^2_{n-1}$ (global null)

All of these are the same theorem applied to different projection matrices.

Common mistakes to avoid

PITFALLS

1. Idempotent but not symmetric

Idempotence alone defines a projection, but the clean orthogonal geometry and χ^2 result use a symmetric projection matrix.

2. Ignoring the mean

$Y^T P Y$ is central χ^2 only when $P\mu = 0$. Otherwise it is noncentral.

3. Assuming independence from a decomposition

A sum $Q_1 + Q_2 = \text{total}$ does not by itself imply independence. Orthogonality / Cochran conditions matter.

4. Treating df as a memorized number

Degrees of freedom are ranks of projection matrices - dimensions of subspaces.

Practice: identify the distribution

PRACTICE

- If P is a rank-4 orthogonal projection and $Z \sim N(0, I)$, what is the distribution of $Z^T P Z$?
- In a full-rank normal linear model with $n = 40$ and $p = 6$, what is the distribution of SSE/σ^2 ?
- For nested models with $p_0 = 3$ and $p_1 = 7$, how many degrees of freedom belong to the extra sum of squares?
- Under what condition is $Y^T P Y / \sigma^2$ central rather than noncentral?
- Why are the extra sum of squares and the full-model SSE independent?

Takeaways

SUMMARY

Quadratic form

A symmetric quadratic form diagonalizes into a weighted sum of squared coordinates.

Projection

For a normal vector, a rank- r orthogonal projection produces χ^2_r .

Cochran's theorem

Orthogonal projection pieces of a standard normal vector give independent chi-squared sums.

Linear models

SSE, SSR, and extra sums of squares are all projection quadratic forms.

Inference

The F distribution appears because we divide independent chi-squared quantities by their ranks.

normality + orthogonal geometry $\implies$ exact finite-sample inference